Safety Case Addendum AURIX™ TC39x BD-Step Microcontroller

This Safety Case Addendum is applicable for the following products:

Product Name	Package/Variants
TC39x BD-Step Microcontroller	*Refer to Changes, Table 1*

About this document

Scope and purpose

This document provides the addendum to the previously released Safety Case Report V1.00.

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Executive Summary

This document describes the changes to the last release of TC39x BC-Step Microcontroller Safety Case Report V1.00. Since last release all received changes were analyzed for applicability to the TC39x BC-Step Microcontroller Safety Case Report. Changes were assessed and necessary modification were carried out as per the standard configuration management practice.

The TC39x BD-Step Safety Case Addendum V1.0 is built on the TC39x BC-Step Safety Case Report V1.00. With release of this addendum, application of TC39x BC-Step Microcontroller Safety Case Report V1.00 shall be considered for TC39x BD-Step together with this addendum unless otherwise noted.

Changes

Changes from Safety Case Report V1.00 to Addendum V1.0

Table 1 **Overview of changes (Safety Case Report V1.00 to Addendum V1.0)**

Safety Case Structure	Changes
Chapter 1	No changes
Chapter 2	Changes in Chapter 2.1 – See details below.
Chapter 3	No changes
Chapter 4	Changes in 4.2 – See details below.
Chapter 5	Changes in Chapter 5.4 – See details below
Chapter 6	No changes
Chapter 7	Changes in Chapter 7 – See details below.
Chapter 8	Changed in Chapter 8 – See details below.
Chapter 9	No changes
Chapter 10	No changes

Details of changes:

Chapter 2.1

This Safety Case Addendum applies to the *TC39x BD-Step Microcontroller* product variants given in below table.

Table 1 TC39x BD-Step Microcontroller

Component	Variants	
Hardware	SAL-TC399XE-256F300S	BD
	SAL-TC399XX-256F300S	BD
	SAL-TC399XP-256F300S	BD
	SAL-TC397XP-256F300S	BD
	SAK-TC399XE-256F300S	BD
	SAK-TC397XE-256F300S	BD
	SAK-TC399XP-256F300S	BD
	SAK-TC399XX-256F300S	BD
	SAK-TC397XP-256F300S	BD
	SAK-TC397XA-256F300S	BD
	SAK-TC397QA-160F300S	BD
	SAK-TC397XX-256F300S	BD
	SAK-TC397QP-192F300S	BD
	SAK-TC397QP-256F300S	BD
	SAK-TC397XZ-256F300S	BD
	SAK-TC397XM-256F300S	BD

Changes

Component	Variants
	SAL-TC397XE-256F300S BD SAK-TC397XT-256F300S BD SAL-TC397QP-192F300S BD SAL-TC397QP-256F300S BD SAL-TC397XZ-256F300S BD SAL-TC397XX-256F300S BD
Software Units	Software Built-in Self-Test (SBST) for CPU and SPU

This Safety Case Addendum extends the claims from the TC39x BC-Step Microcontroller Safety Case Report V1.00 to the variants listed in the table above.

Chapter 4.2

This section shows the additional top level safety requirement (TLSR) which is applicable for the TC39x BD-Step Microcontroller. The TLSRs listed in the TC39x BC-Step Microcontroller Safety Case Report V1.00 remains valid for TC39x BD-Step.

4.2.25 [MCU-TLSR] Digital Acquisition [ASIL B]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Digital Acquisition] [MCU Function].

Chapter 5.4

Table 5 Performed Audit and Assessments

Type	Scope	Start Date	End Date	Status	Rating
Design Assessment (ISO26262)	*SBST (CPU)*	15/07/2019	15/07/2019	Conducted	Pass
Verification Assessment (ISO26262)	*SBST (CPU)*	15/07/2019	15/07/2019	Conducted	Pass
Production Release Assessment including relevant aspects for Requirements	*TC39x BD-Step Microcontroller & SBST*	25/03/2020	26/03/2020	Conducted	Pass

Changes

and Safety Concept Assessment and Verification Assessment (ISO26262)					

Chapter 7

Table 8 in the Safety Case Report V1.00 remains unchanged even though the Work Products and Internal Documents are newly baselined according to the following internal baseline: 6247

Change with new baseline information. Table 9 with customer relevant documentation is applicable for all product variants covered in this Safety Case Addendum.

Changes

Table 9 Reference List of Customer Documentation and Internal Documents

Document	Customer Version	Release Date	Internal Document
AURIX™ TC3xx IFX User Manual	V1.2.0	2019-04	MC_ACE_PRJDOC_UM
AURIX™ TC39x Appendix to User's Manual	V1.2.0	2019-04	MC_ACE_PRJDOC_UM
AURIX™ TC3xx ED Target Specification[1]	V2.8.0	2019-05	MC_ACE_PRJDOC_UM
TC39xB_DS_v11	V1.1	2019-09	MC_ACE_PRJDOC_DS
TC39x_Data_Sheet_Addendum_v1.4	V1.4	2020-01	MC_ACE_PRJDOC_DS
AURIX TC3xx Safety Manual v1.08	V1.08	2020-03	MC_ACE_PRJDOC_SM
TC39xBD_Errata_Sheet_v1_0	V1.0	2019-12-19	MC_ACE_PRJDOC_ES
AURIX_TC39x_FMEDA_v1.01_ext	V1.01_ext	2018-12-17	MC_PRJDOC_FMEDA
CPU SBST User Manual	V1.0.3.1.0[2]	2019-03-05	MC_PRJDOC_SWDI_SBST_CPU
SPU SBST User Manual	V1.0.1.1.0	2019-02-11	MC_PRJDOC_SWDI_SBST_SPU
A2G_AURIX Safety Analysis Summary Report	V1.12	2020-03-10	MC_PRJDOC_SASR
TC39xBD_SafetyCaseAddendum_V1.0	V1.0	2020-03	MC_PRJDOC_SCR
TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	V1.2.2	2020-01-15	Z8F54561158
TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	V1.2.2	2020-01-15	Z8F54351040

[1] Distribution under NDA, only relevant for tool development not for application development

[2] Customers can still use the previous version (V1.0.2.1.0). IFX highly recommends to use the version stated in the table for ongoing and future developments.

Changes

Chapter 8

The following table is applicable for TC39xBD and shall be carefully read and assessed by all relevant stakeholders. IFX assumes that the information contained in this table is forwarded to all relevant parties within the organization of the recipient of this document.

Table 10 Open Problems List

No.	Open Problem Description	Argument for Acceptance
1	**EVRC Frequency and Phase Synchronization to CCU6/GTM Input** As documented in section "EVRC Frequency and Phase Synchronization to CCU6/GTM Input" in TC3xx UM V1.x.0: • Loss of Synchronization Lock event is triggered if switching frequency range is violated. Loss of Synchronization Lock event is indicated via status bits and interrupt. If this functionality is used in a safety related application, the faults leading to loss of synchronization could be detected by reading the status bit or reacting on the interrupt. This ESM is not part of the Safety Manual.	Current FMEDA provides FIT rate for faults of this functionality if configured to be used in safety related application. This is such small that FIT targets are not violated.
2	Customer documentation (FMEDA) does not provide evidence for usage of GPT12 in safety related applications in combination with GTM. However, Safety Manual still shows ESM which refer to GPT12. It is still possible for the integrator to use GPT12 upon own assessment.	Current FMEDA provides FIT rate for faults of GPT12 functionality if used in safety related application. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
3	Customer documentation (FMEDA) does not provide evidence for usage of PORTS as single channel in safety related applications. It is still possible for the integrator to use PORTS as single channel upon own assessment.	Current FMEDA provides FIT rate for faults of PORTS functionality if used as single channel in safety related applications. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
4	Safety Case Report 1.0 is baselined to a requirement database which has minor number of known incorrect relations of safety requirements (< 1%).	Defendable, all incorrect relations analyzed and assessed, no safety related issues found
5	Risk of input degradation under humidity/temperature cycling stress	Comprehensive safety impact analysis provides evidence, that occurrence of observed fail in the field will not compromise the product safety related aspects.
6	The description of the Safety mechanisms ESM[SW]:EDSADC:VAREF_PLAUSIBILITY and ESM[SW]:EVADC:VAREF_PLAUSIBILITY require customers to implement a check of the ADC reference voltage (VAREF / VAGND). However, it is not explicitly stated, that the result of the check depends on the discharging conditions at VAREF (e.g. number of conversions at a time).	It is assumed that customer implements VAREF plausibility check under representative application conditions
7	Weakness in user documentation regarding "EDSADC Filter Chain Integrator operation – result register reading". Current documentation is not clear regarding validity of the result register	To overcome the corner case, application shall use "Result Handling Via FIFO" only. In this case, results are stored in FIFO and available to the application via read of RESMx register. Reading of

Changes

No.	Open Problem Description	Argument for Acceptance
	content after the integration has stopped. In certain corner case result register may get overwritten before application reads it.	the RESMx can be configured via RFCx and DICFGx.DRM as per application scheduling.
8	Safety mechanism ESM[SW]:SPU:SBST is listed in in FMEDA sheet "Safety Mechanisms" with a wrong D.C. value. Safety mechanism ESM[SW]:SPU:SBST should not be listed there.	Safety mechanism ESM[SW]:SPU:SBST is not used for the FMEDA target metrics evaluation. FMEDA results as documented in SAR are correct. Customer as well is not intended to use safety mechanism ESM[SW]:SPU:SBST. Instead safety mechanism SM[SW]:SPU:SBST is listed in FMEDA sheet "Safety Mechanisms" with correct D.C. value and is used in FMEDA correctly.

Revision history

Document version	Date of release	Description of changes
V1.0	2020-03	Released

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V1.0

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